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CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

CHEMISTRY

Paper 02 – General Proficiency

1 hour 45 minutes

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **There are FIVE questions in this booklet. Answer ALL questions.**
2. **You MUST use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.**
3. **Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.**
4. **The use of non-programmable calculators is allowed.**

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Answer ALL questions.

Do NOT spend more than 30 minutes on Question 1.

1. (a) A student is asked to make a solution of sodium hydroxide by dissolving an unknown mass of the pellets in 250 cm^3 of water. 25.0 cm^3 of the solution is then titrated against a standard solution of hydrochloric acid (0.050 mol dm^{-3}).
Figure 1 below shows the initial and final volumes of the burette reading for the volume of acid added.

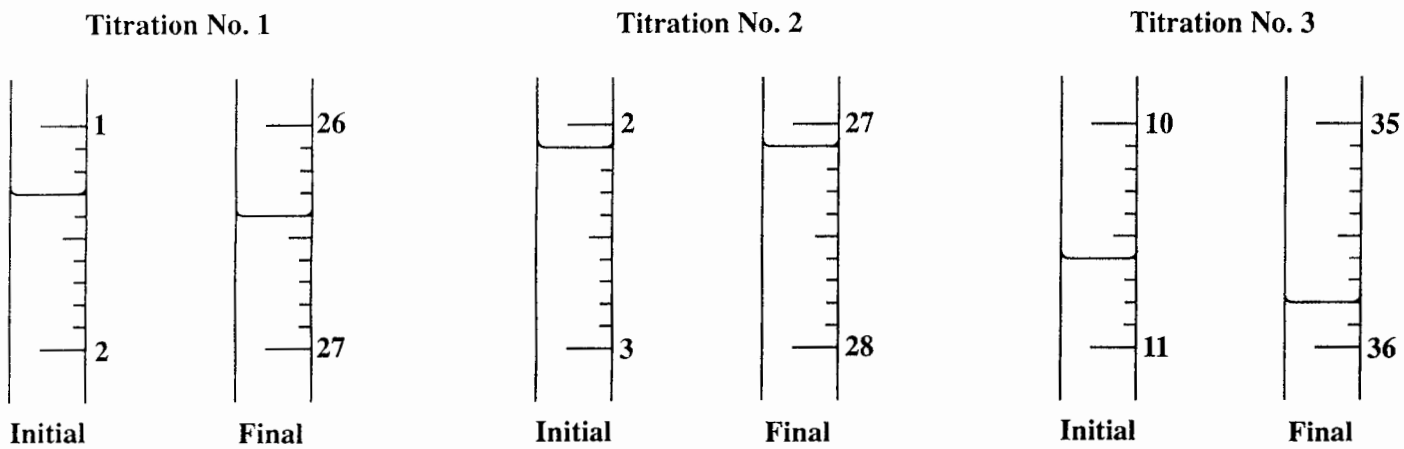

Figure 1. Burette readings showing volumes of acid used

(i) Use the information given in Figure 1 above to complete Table 1 below.

TABLE 1: RESULTS OF EXPERIMENT

	Titration number		
	1	2	3
Final volume / cm^3			
Initial volume / cm^3			
Volume used / cm^3			

Average volume of acid used / cm^3

_____

(4 marks)

(ii) Write a chemical equation for the reaction that occurs during the titration.

_____

(1 mark)

From the data in Table 1 calculate EACH of the following:

(iii) The number of moles of hydrochloric acid used in the titration

(1 mark)

(iv) The number of moles of sodium hydroxide in 25.0 cm^3 of the solution

(1 mark)

(v) The number of moles of sodium hydroxide in 250 cm^3 of solution

(1 mark)

(vi) Mass of sodium hydroxide dissolved in 250 cm^3 of water

(Relative atomic mass: Na = 23, O = 16, H = 1)

(2 marks)

(b) A student carries out the following tests on a solid, Q, and makes the observations recorded in Table 2 below. Complete Table 2 to show all possible inferences and write ionic equations where required.

TABLE 2: RESULTS OF TESTS ON SOLID Q

Tests	Observations	Inferences
(i) A small amount of Q is heated in a dry test tube.	• A brown gas evolves which turns moist blue litmus red. • Another gas evolves which rekindles a glowing splint.	• •
A portion of Q is dissolved in approximately 10 cm^3 of deionized water. The resulting solution is divided into 3 portions for Tests (ii) - (iv).		
(ii) To one portion of Solution Q from above, aqueous sodium hydroxide is added dropwise until in excess.	• A white precipitate is formed • Precipitate is soluble in excess sodium hydroxide.	• •
(iii) To another portion of Q from above, aqueous ammonia is added dropwise until in excess.	• A white precipitate is formed • Precipitate is insoluble in excess aqueous ammonia.	• •
(iv) To another portion of Q from above, aqueous potassium iodide is added.	• A yellow precipitate is formed.	• (Ionic equation required)

(7 marks)

(v) Suggest the possible identity of Q.

_____

(1 mark)

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(c) You are provided with a bottle of sea water which also contains some sand.

Plan and design an experiment to recover the solid 'sea salt' from the mixture. Your answer should include the following:

(i) A suggested list of apparatus which you would use in recovering the solid sea salt from the sandy sea water

(4 marks)

(ii) An outline of the steps for the procedure you could use

(4 marks)

(iii) List the main observations that would be expected at each stage of the experiment.

(2 marks)

(iv) A student suggests that solid sea salt contains chloride ions. How would you go about testing for chloride ions in the solid sea salt?

(2 marks)

Total 30 marks

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2. Table 3 presents some information on the reactions of Compound A and copper metal with various reagents. Use the information in Table 3 to answer the questions that follow.

TABLE 3: SOME REACTIONS OF COMPOUND A AND COPPER

	Reactions with		
	Acidified potassium chromate (VI) solution	Aqueous iron (II) nitrate	Aqueous silver nitrate solution
Aqueous solution of A	Experiment 1 Potassium chromate (VI) changes from orange to green.	Experiment 2 The iron (II) nitrate changes from pale green to pale yellow.	
Copper metal		Experiment 3	Experiment 4 Silver solid is deposited. Pale blue solution is formed.

(a) (i) Classify Solution A as an oxidizing or a reducing agent in Experiments 1 and 2.

Experiment 1 : A is acting as . . . . . .

Experiment 2 : A is acting as . . . . . .

(2 marks)

(ii) Using the information given in Table 3 only, explain your answer to (a) (i) above.

Experiment 1:

_____

_____

Experiment 2:

_____

_____

(4 marks)

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(b) (i) Write a balanced ionic equation to represent the reaction that occurs in Experiment 4.

_____

(2 marks)

(ii) Account for the pale blue solution that is observed in Experiment 4.

_____

(1 mark)

(c) With reference to the electrochemical series, predict what you may be expected to observe in Experiment 3. Give a reason for your answer.

_____

_____

_____

_____

(3 marks)

(d) When exposed to air for some time, copper ornaments become tarnished by green copper carbonate.

(i) Suggest the name of a common household chemical that can be used to clean such tarnished copper ornaments.

_____

(1 mark)

(ii) Write a suitable ionic equation to show the reaction in (d) (i) above.

_____

(2 marks)

Total 15 marks

3. The information provided in Table 4 below refers to atoms of two elements, A and B.

(a) Complete Table 4 by writing the required information in the relevant spaces.

TABLE 4: ATOMIC PROPERTIES OF ELEMENTS A AND B

	A	B
Mass number		19
Atomic number	12	
Number of protons		9
Number of electrons	12	9
Number of neutrons	12	
Electronic configuration		
Group number	2	
Period number		2

(4 marks)

(b) (i) What type of bonding would take place between A and B?

_____

(1 mark)

(ii) Give a reason for your answer in (b) (i) above.

_____

_____

_____

(2 marks)

(iii) Use dot cross diagrams to illustrate the bonding between A and B.

(2 marks)

(c) Describe the bonding expected in A.

(2 marks)

(d) (i) Describe the bonding in graphite.

(2 marks)

(ii) Graphite is used as a lubricant. How does this use relate to its structure and bonding?

(2 marks)

Total 15 marks

4. X and Y are two hydrocarbons. When X is completely burnt in air, 0.50 moles of X produce 60 dm^3 of CO_2 and 3 moles of water at r.t.p.

(a) Deduce the molecular formula of X. (1 mole of gas at r.t.p. has a volume of 24 dm^3.)

(4 marks)

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(b) (i) Based on your answer in 4 (a) on page 9, deduce the homologous series to which X belongs.

_____

(1 mark)

(ii) Write the name of Compound X.

_____

(1 mark)

(iii) Draw the fully displayed structure of X.

(1 mark)

(c) Y has the molecular formula, C_5H_{10}.

(i) To what homologous series does Y belong?

_____

(1 mark)

(ii) Write the name of Compound Y.

_____

(1 mark)

(iii) Draw the fully displayed structure of Y.

(2 marks)

(d) What chemical test may be used to distinguish between Compounds X and Y?

Reagent: _____

Observation: _____

_____

(2 marks)

(e) What reagent and reaction conditions are necessary to convert Y to X?

Reagent: _____

Conditions: _____

(2 marks)

Total 15 marks

5. (a) Define EACH of the following terms:

(i) Amphoteric oxide

_____

_____

(2 marks)

(ii) Acid anhydride

_____

_____

(2 marks)

(b) Describe appropriate chemical tests, the results of which could be used to determine whether zinc oxide is amphoteric or whether it is an acid anhydride.

Chemical tests:

(i) To determine if zinc oxide is amphoteric

_____

_____

_____

(ii) To determine if zinc oxide is an acid anhydride

_____

_____

_____

(5 marks)

(c) Sulphur (IV) oxide, a gaseous pollutant, dissolves in water droplets in the atmosphere and thus causes acid rain.

(i) Write the formula for sulphur (IV) oxide.

_____

(1 mark)

(ii) Write an equation for the reaction between sulphur (IV) oxide and water.

_____

(1 mark)

(d) It has been suggested that the amount of polluting gases emitted into the atmosphere, such as the oxides of sulphur and nitrogen which are acidic, could be reduced by first passing exhaust air from factories through beds of powdered reagents.

Based on this information:

(i) Give the name of ONE reagent that could be used for this purpose.

Support your answer by writing a chemical equation, using ONE of the named pollutants mentioned above.

_____

_____

(2 marks)

(ii) Suggest why the reagent you have named in (d) (i) above should be used in the powdered form.

_____

_____

(2 marks)

Total 15 marks

END OF TEST